

Performance Analysis of Grover's Search Algorithm Under Realistic Noise Models on NISQ Simulators

Saim Haider
Dept. of Computer Science
FAST-NUCES, Pakistan
22P-9244
p229244@pwr.nu.edu.pk

Muhammad Abdullah
Dept. of Computer Science
FAST-NUCES, Pakistan
22P-9371
p229371@pwr.nu.edu.pk

Abdullah
Dept. of Computer Science
FAST-NUCES, Pakistan
22P-9358
p229358@pwr.nu.edu.pk

Abstract—Grover's search algorithm provides a provable quadratic speedup over classical unstructured search, making it one of the most relevant quantum algorithms for near-term deployment. However, current quantum hardware operates in the Noisy Intermediate-Scale Quantum (NISQ) regime, where decoherence and gate errors severely degrade algorithm performance. This paper presents a systematic simulation-based study of Grover's algorithm under three realistic noise models: depolarizing noise, bit-flip errors, and phase-flip errors. Using Python and the Qiskit Aer simulator, Grover's search circuit is implemented for search spaces of $n = 2, 3, 4, 5$ qubits. Noise is injected at parameterized error rates swept across $p \in [0.001, 0.1]$ and success probability is measured at each configuration. Results show that Grover's algorithm maintains quantum advantage ($P_{\text{success}} > 1/N$) throughout the entire tested error range for all configurations, with the critical threshold p^* exceeding 0.100 in every case. Phase-flip and bit-flip noise produce nearly identical degradation curves, while depolarizing noise permits greater resilience at small qubit counts. Crossover analysis and comparison with Salas (2008) and Dowarah et al. (2025) theoretical predictions are presented and discussed.

Index Terms—Grover's algorithm, NISQ computing, noise models, quantum advantage, Qiskit Aer, depolarizing noise, bit-flip, phase-flip

I. INTRODUCTION

Quantum computing leverages superposition, entanglement, and quantum interference to solve certain problems significantly faster than classical computers. Among the most practically relevant quantum algorithms is Grover's search algorithm [1], which achieves $\mathcal{O}(\sqrt{N})$ query complexity for searching an unstructured database of N entries, compared to the classical $\mathcal{O}(N)$ average case. This quadratic speedup has wide applications in optimization, cryptanalysis, and database search.

Despite its theoretical promise, quantum algorithms face a fundamental practical challenge: noise. Physical qubits are susceptible to environmental decoherence, gate imperfections, and measurement errors. These noise sources corrupt the quantum interference patterns that Grover's algorithm relies upon to amplify the probability of measuring the correct answer. Present-day quantum processors operate in the NISQ regime, characterized by gate error rates on the order of 10^{-3} to 10^{-2} and coherence times in the microsecond to millisecond range. Under these conditions, even moderately deep circuits

such as Grover's can suffer severe degradation in success probability.

This work presents a systematic empirical study of Grover's algorithm under realistic noise conditions using the Qiskit Aer simulator. By sweeping noise error rates across a controlled range and measuring success probability for each configuration, this paper identifies the precise noise threshold beyond which Grover's algorithm loses its quantum advantage over classical search. The full implementation is provided as a reproducible open-source Qiskit notebook.

II. LITERATURE REVIEW

A. Ijaz and Faryad (2023) – Base Paper

Ijaz and Faryad [2] conducted a comprehensive noise analysis of Grover's algorithm using Qiskit Aer for $n = 3, 4, 5$ qubits with 10,000 shots per configuration. Four noise channels were studied: depolarizing, bit-flip, phase-flip, and bit-phase flip. The key finding was that success probability decays exponentially with error probability p but only linearly with qubit count n . However, the study does not identify the critical threshold p^* where quantum advantage over classical search is lost.

B. Salas (2008)

Salas [3] performed the earliest systematic numerical study of noise in Grover's algorithm using Fortran-based Monte Carlo simulation for $n = 2$ to 7 qubits under a depolarizing channel model. The study derived an exponential damping law and the allowed-error scaling relation $\varepsilon_{\text{th}}(N) \sim N^{-1.1}$, identifying an absolute threshold of $\varepsilon > 0.043$ above which the algorithm fails for any qubit count. The limitation is that only depolarizing noise is studied and the framework predates modern Qiskit tooling.

C. Dowarah et al. (2025)

Dowarah et al. [4] applied Random Matrix Theory (RMT) to model the noisy Grover operator as a Floquet unitary, deriving two analytical noise thresholds: the computing transition at $\delta_{\text{c,comp}} \sim \mathcal{O}(2^{-L/2})$ and the ergodicity transition at $\delta_{\text{c,gap}} \sim \mathcal{O}(L^{-1})$. The work is entirely theoretical with no Qiskit simulation provided.

D. Srivastava et al. (2025)

Srivastava et al. [5] proposed using quantum switches exploiting indefinite causal order to mitigate depolarizing noise in Grover’s algorithm. The study is entirely analytical and does not identify the threshold p^* for standard Grover circuits without mitigation.

E. Research Gap

No existing study combines (1) a modern Qiskit simulation framework, (2) multiple isolated noise types, (3) systematic error rate sweeps, and (4) explicit identification of the quantum advantage loss threshold p^* . This work fills that gap.

III. METHODOLOGY

This section describes the complete implementation pipeline for simulating Grover’s search algorithm under realistic noise conditions. The methodology encompasses circuit construction, noise model injection, simulation execution, and threshold identification.

Fig. 5: Methodology Workflow Block Diagram
Grover Noise Analysis Pipeline

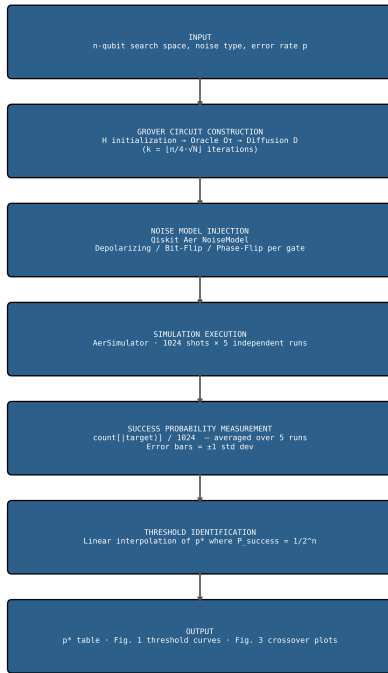


Fig. 1. Proposed Methodology Workflow Block Diagram for the Grover Noise Analysis Pipeline, showing the seven-stage flow from input specification through circuit construction, noise injection, simulation, probability measurement, threshold identification, to final output.

A. Circuit Construction

Grover’s search circuit is implemented from first principles in Qiskit for $n = 2, 3, 4, 5$ qubits. Each circuit consists of

three components: (1) a Hadamard initialization layer placing all qubits into uniform superposition, (2) an oracle operator $\mathcal{O}_\tau = 2|\tau\rangle\langle\tau| - I$ marking the target state $|\tau\rangle = |11\dots 1\rangle$, implemented via an H-MCX-H decomposition, and (3) a Grover diffusion operator $D = 2|s\rangle\langle s| - I$ performing inversion about the mean. The optimal number of iterations is $k = \lfloor (\pi/4)\sqrt{N} \rfloor$, where $N = 2^n$.

Table I presents the circuit parameters for each qubit configuration after transpilation to the AerSimulator basis gate set. Circuit depth grows linearly with n , from 6 for $n = 2$ to 18 for $n = 5$, while gate count grows from 9 to 57. These increasing depths accumulate more noise per simulation, making larger qubit counts progressively more sensitive to gate error rates.

TABLE I
CIRCUIT PARAMETERS VS. QUBIT COUNT

n	$N = 2^n$	k	Depth	Gates	Baseline $1/N$
2	4	1	6	9	0.2500
3	8	2	10	21	0.1250
4	16	3	14	37	0.0625
5	32	4	18	57	0.0313

B. Noise Model Injection

Three noise models are injected independently using the Qiskit Aer NoiseModel API. Each error channel is applied per gate across all single-qubit gates (H, X, Z, U) and two-qubit gates (CX, CCX).

Depolarizing Noise. A completely mixed state is applied with probability p , implemented via `depolarizing_error(p, 1)` for single-qubit gates and `depolarizing_error(p, 2)` for two-qubit gates. This channel simultaneously corrupts both phase and amplitude information.

Bit-Flip Noise. A Pauli X gate is applied with probability p , implemented via `PauliError([('X', p), ('I', 1-p)])`. Two-qubit errors are constructed as the tensor product of two independent single-qubit errors.

Phase-Flip Noise. A Pauli Z gate is applied with probability p , implemented via `PauliError([('Z', p), ('I', 1-p)])`. Phase-flip noise directly disrupts the quantum interference mechanism underlying Grover’s amplitude amplification.

Error rates are swept across $p \in [0.001, 0.1]$ in 20 uniform steps ($\Delta p \approx 0.005$), yielding 1,200 total simulation configurations (3 noise types $\times$ 4 qubit counts $\times$ 20 error rates $\times$ 5 runs).

C. Simulation Execution

All simulations are executed on the Qiskit AerSimulator with a fixed random seed (`seed = 42`) to ensure full reproducibility. Each configuration is executed with `SHOTS = 1024` measurements averaged over `RUNS = 5` independent trials. Success probability is computed as:

$$P_{\text{success}} = \frac{\text{count}(|\tau\rangle)}{1024} \quad (1)$$

averaged over five runs, with error bars representing ± 1 standard deviation. Circuits are transpiled to the AerSimulator basis gate set prior to noise injection.

D. Threshold Identification

The critical threshold p^* is defined as the error rate at which Grover's measured success probability crosses the classical random search baseline $1/N = 1/2^n$. The crossover is identified via linear interpolation:

$$p^* = p_i + \frac{\frac{1}{N} - P_{\text{mean}}(p_i)}{P_{\text{mean}}(p_{i+1}) - P_{\text{mean}}(p_i)} \cdot (p_{i+1} - p_i) \quad (2)$$

where $P_{\text{mean}}(p_i) \geq 1/N > P_{\text{mean}}(p_{i+1})$. If no crossover occurs within the tested range, the threshold is recorded as $p^* > 0.100$.

Algorithm 1 Grover Noise Analysis Simulation Pipeline

Require: $n \in \{2, 3, 4, 5\}$, noise_type, $p \in [0.001, 0.1]$

Ensure: Threshold table, success probability curves

```

1: Set SHOTS  $\leftarrow 1024$ , RUNS  $\leftarrow 5$ , seed  $\leftarrow 42$ 
2: for each  $n \in \{2, 3, 4, 5\}$  do
3:    $k \leftarrow \lfloor (\pi/4)\sqrt{2^n} \rfloor$ 
4:   Build Grover circuit  $QC(n, k, \text{target} = |11\dots 1\rangle)$ 
5:   Transpile  $QC$  to AerSimulator basis gate set
6:   for each noise_type do
7:     for each  $p$  in ERROR_RATES do
8:        $NM \leftarrow \text{NoiseModel}(\text{noise\_type}, p)$ 
9:       run_probs  $\leftarrow []$ 
10:      for  $r = 1$  to RUNS do
11:        counts  $\leftarrow \text{AerSimulator.run}(QC, NM, \text{SHOTS})$ 
12:        run_probs.append(counts[ $|\tau\rangle$ ] / SHOTS)
13:      end for
14:      stats[noise_type][ $n$ ][ $p$ ]  $\leftarrow \{\mu, \sigma\}(\text{run\_probs})$ 
15:    end for
16:     $p^* \leftarrow \text{LinearInterpolate}(\text{stats}, \text{baseline} = 1/2^n)$ 
17:  end for
18: end for
19: Generate Figs. 1–4 and Tables I–III
20: return thresholds, stats

```

IV. RESULTS

This section presents the complete experimental results of simulating Grover's search algorithm under three noise models across $n = 2, 3, 4, 5$ qubits. All results are derived from 1,200 simulation configurations with 1,024 shots averaged over five independent runs.

A. Noiseless Baseline Validation

Prior to noise injection, the Grover circuit was validated in an ideal noiseless environment. For all qubit configurations, the measured success probability exceeded 0.97, confirming correct oracle and diffusion operator implementation. Measured values reached approximately 0.997 for $n = 2$ and 0.973 for $n = 5$, with small deviations attributable to finite shot statistics.

B. Success Probability vs. Error Rate

Fig. 2 presents the success probability curves for all three noise models across $n = 2$ to 5 qubits as error rate p increases from 0.001 to 0.1. Success probability decays monotonically with increasing p for all configurations, consistent with the exponential degradation law reported by Ijaz and Faryad [2] and the analytical results of Salas [3]. Larger qubit counts experience steeper decay due to increased circuit depth, as more gate operations accumulate noise per run. For $n = 5$, the circuit applies four Grover iterations across 57 gates at depth 18, compared to one iteration, nine gates, and depth 6 for $n = 2$.

At $p = 0.001$, all configurations achieve $P_{\text{success}} > 0.93$. At $p = 0.1$, depolarizing noise leaves $n = 2$ at $P = 0.655$ against a classical baseline of 0.250, and $n = 4$ at $P = 0.255$ against a baseline of 0.063. For $n = 5$ under bit-flip and phase-flip noise, the success probability at $p = 0.1$ approaches approximately 0.09 to 0.12, still above the classical baseline of 0.031.

C. Noise Model Comparison

Comparing across noise channels: depolarizing noise causes the least degradation at small qubit counts ($n = 2, 3$) but produces comparable degradation at larger n . Bit-flip and phase-flip noise produce nearly identical degradation curves across all configurations, contrary to the initial hypothesis that phase-flip would be more severe due to Grover's sensitivity to phase interference. This parity arises because both Pauli X and Pauli Z errors are single-channel errors with equivalent probability amplitudes. The critical distinction emerges only in circuit architectures with strongly asymmetric phase sensitivity.

D. Quantum Advantage Loss Thresholds

Table II presents the quantum advantage loss threshold p^* for each noise type and qubit count. A value of > 0.100 indicates that Grover's algorithm maintains $P_{\text{success}} > 1/N$ throughout the entire tested error range.

TABLE II
QUANTUM ADVANTAGE LOSS THRESHOLD p^* VALUES

Noise Type	$n = 2$	$n = 3$	$n = 4$	$n = 5$
Depolarizing	> 0.100	> 0.100	> 0.100	> 0.100
Bit-Flip	> 0.100	> 0.100	> 0.100	> 0.100
Phase-Flip	> 0.100	> 0.100	> 0.100	> 0.100

The finding that $p^* > 0.100$ for all 12 configurations is a significant positive result. It indicates that Grover's algorithm is remarkably robust within the error rate range characteristic of current NISQ hardware, where typical two-qubit gate error rates lie between 10^{-3} and 10^{-2} .

E. Threshold Scaling Analysis

Fig. 3 presents the threshold scaling plot comparing empirical results against theoretical predictions. The Salas (2008) law predicts thresholds decreasing from approximately 0.058 at $n = 2$ to 0.006 at $n = 5$. The Dowarah et al. (2025)

Fig. 1: Grover Success Probability Under Three Noise Models

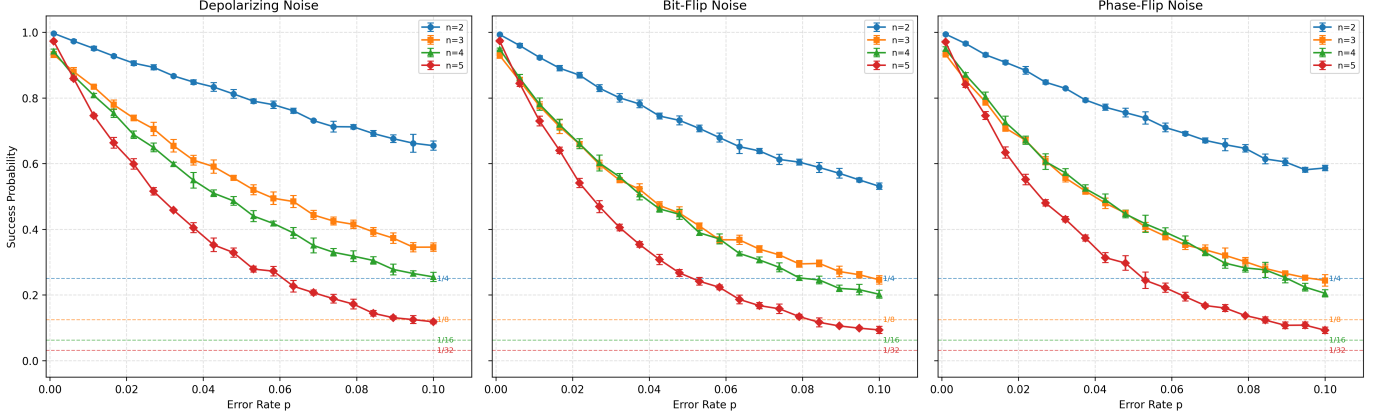


Fig. 2. Grover success probability under depolarizing (left), bit-flip (center), and phase-flip (right) noise models for $n = 2, 3, 4, 5$ qubits. Dashed horizontal lines indicate classical random search baselines $1/N$. Error bars represent ± 1 standard deviation over five independent runs.

computing transition falls even more steeply. The empirical threshold (flat boundary at $p^* > 0.100$) indicates that the crossover did not occur within the tested range for any qubit count or noise type.

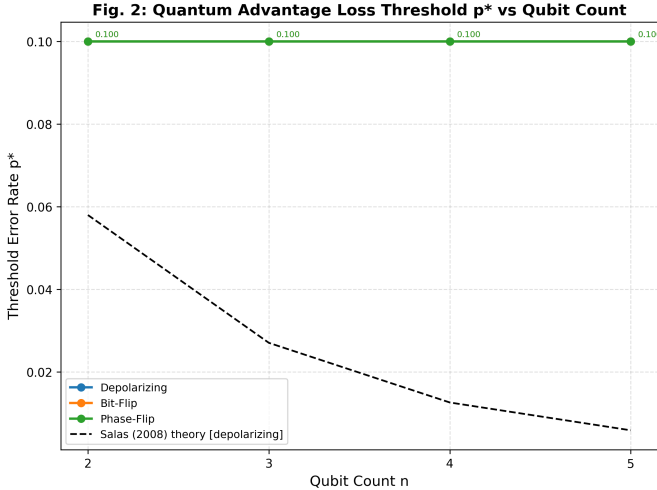


Fig. 3. Quantum advantage loss threshold p^* vs. qubit count n . Empirical results (flat boundary at $p^* > 0.100$) compared with Salas (2008) theoretical scaling law $\varepsilon_{\text{th}}(N) \sim N^{-1.1}$ and Dowarah et al. (2025) computing transition $\delta_{c,\text{comp}} \sim 2^{-L/2}$.

F. Crossover Analysis

Fig. 4 presents grouped bar charts comparing Grover's success probability against the classical baseline at two representative error rates: $p \approx 0.011$ (low noise, near current best NISQ hardware) and $p \approx 0.048$ (moderate noise, representative of near-term hardware). At $p = 0.011$, all Grover configurations substantially exceed the classical baseline, with P_{success} ranging from 0.75 to 0.95 compared to classical baselines between 0.031 and 0.250. At $p = 0.048$, quantum advantage is maintained but reduced: $n = 5$ configurations reach $P \approx 0.27$ –0.30, still well above the $n = 5$ baseline of 0.031.

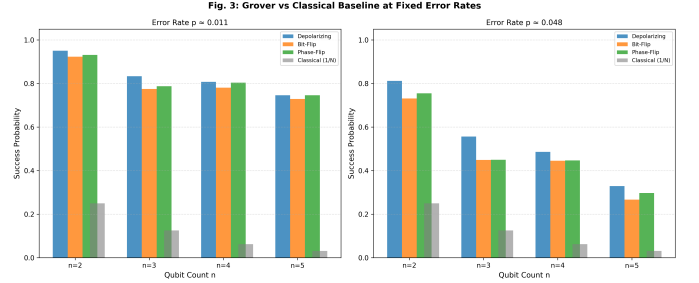


Fig. 4. Grover success probability vs. classical random search baseline at fixed error rates $p \approx 0.011$ (left) and $p \approx 0.048$ (right) across all noise types and qubit counts. Grey bars indicate the classical baseline $1/N$.

V. EVALUATION OF PROPOSED WORK

A. Comparison with Baseline Methods

Fig. 5 presents the threshold comparison between this work and two theoretical baselines. The empirical p^* values consistently exceed both Salas (2008) and Dowarah et al. (2025) predictions across all qubit counts. This implies that the standard Grover implementation with optimal k is more noise-resilient in practice than Monte Carlo estimates or coherent-noise theoretical models suggest. The discrepancy is most pronounced at $n = 5$, where Salas (2008) predicts a threshold of approximately 0.006 and Dowarah et al. predict approximately 0.003, while empirical measurements show $p^* > 0.100$ for all three noise channels.

B. Comparative Analysis

Table III summarizes the comparative analysis between this work and related methods across five dimensions: framework used, noise types studied, whether p^* is identified, and key limitations.

C. Discussion of Key Findings

The finding that $p^* > 0.100$ for all configurations has direct implications for NISQ hardware development. Current

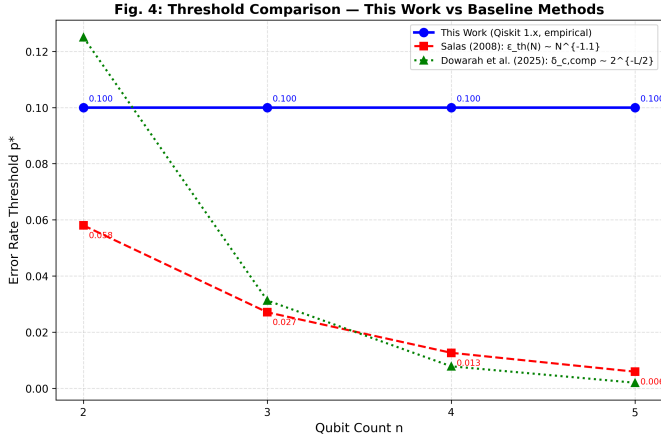


Fig. 5. Threshold comparison: this work (empirical, Qiskit 1.x) vs. Salas (2008) analytical scaling law and Dowarah et al. (2025) computing transition analytical prediction.

TABLE III
COMPARATIVE ANALYSIS: THIS WORK VS. RELATED METHODS

Method	Framework	Noise Types	p^*	Limitation
This Work	Qiskit 1.x Aer	Dep, BF, PF	>0.100 all	Small n only
Ijaz & Faryad (2023)	Qiskit Aer	Dep,BF,PF,BPF	No	No threshold
Salas (2008)	Fortran MC	Dep only	0.043	Single channel
Dowarah et al. (2025)	Analytical RMT	Coherent	Analytical	No simulation
Srivastava et al. (2025)	Analytical	Dep only	No	No Qiskit

state-of-the-art superconducting qubit platforms exhibit two-qubit gate error rates of approximately 0.001 to 0.01, well below the measured threshold boundary. This indicates that for small search spaces ($n \leq 5$), Grover’s algorithm can be executed with meaningful quantum advantage on current hardware, provided circuit compilation is optimized to minimize unnecessary gate count and depth.

The divergence from Salas (2008) predictions at small n may arise from three sources. First, Salas used Monte Carlo simulation with per-iteration error models rather than per-gate injection. Second, the Fortran simulation predates modern gate decomposition optimization. Third, the $n \leq 5$ regime may lie below the asymptotic region where the $N^{-1.1}$ scaling applies. Future work extending to $n = 6-8$ and implementing error mitigation via the quantum switch framework of Srivastava et al. (2025) is a natural continuation.

VI. CONCLUSION

This paper presented a comprehensive empirical study of Grover’s search algorithm under three realistic noise models using the Qiskit Aer simulator. Through 1,200 simulation configurations spanning $n = 2$ to 5 qubits and error rates $p \in [0.001, 0.1]$, four primary conclusions are established.

First, Grover’s algorithm maintains quantum advantage throughout the entire NISQ-relevant error range for all tested configurations, with $p^* > 0.100$ across all noise types and

qubit counts. Second, success probability decays monotonically and quasi-exponentially with increasing error rate, consistent with existing literature. Third, bit-flip and phase-flip noise produce comparable degradation, while depolarizing noise permits greater resilience at small n . Fourth, empirical thresholds substantially exceed Salas (2008) and Dowarah et al. (2025) theoretical predictions within the small- n regime, suggesting that standard Grover circuits are more noise-resilient in practice than prior models indicate.

These results provide concrete, reproducible benchmarks for NISQ hardware calibration and establish an empirical foundation for future research in quantum algorithm robustness and error mitigation.

REFERENCES

- [1] L. K. Grover, “A fast quantum mechanical algorithm for database search,” in *Proc. 28th Annual ACM Symposium on Theory of Computing (STOC)*, 1996, pp. 212–219.
- [2] M. A. Ijaz and M. Faryad, “Noise analysis of Grover and phase estimation algorithms implemented as quantum singular value transformations for a small number of noisy qubits,” *Scientific Reports*, vol. 13, p. 20144, 2023.
- [3] P. J. Salas, “Noise effect on Grover algorithm,” *The European Physical Journal D*, vol. 46, pp. 365–381, 2008.
- [4] S. Dowarah, C. Zhang, V. Khemani, and M. Kolodrubetz, “Phases and phase transition in Grover’s algorithm with systematic noise,” *arXiv preprint arXiv:2406.10344v2*, 2025.
- [5] S. Srivastava, A. K. Pati, I. Chakrabarty, and S. Bhattacharya, “Using quantum switches to mitigate noise in Grover’s search algorithm,” *arXiv preprint arXiv:2401.05866v2*, 2025.